Jake Brawer, Ph.D.

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Research overview

Summary I build **human–robot collaborative systems** that learn through their interactions. My research draws upon recent innovations in machine learning as well as foundational techniques in artificial intelligence for designing systems that learn and utilize **abstract knowledge** to make social and physical interactions more **naturalistic** and **robust**.

Education

2023-Current **Postdoctoral Researcher** University of Colorado, Boulder
Advisors: Alessandro Roncone and Bradley Hayes

2016-2023 **Ph.D. in Computer Science** Yale University, Advisor: Brian Scassellati

2012-2016 **B.A. in Cognitive Science; Computer Science minor** Vassar College

In progress

Debasmita Ghose, Oz Gitelson, Michal Lewkowicz, Jake Brawer, Marynel Vazquez, and Brian Scassellati (2025). "Robots that Reveal Humans' Goals Using Critical Decision Points During Collaboration". In: *Proceedings of the Robotics: Science and Systems*. Los Angeles, California.

Journal articles

Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022a). "**Robot Tool Use: A Survey**". In: *Frontiers in Robotics and AI* 9, p. 369.

* Meiying Qin, * **Jake Brawer**, and Brian Scassellati (2021). "**Rapidly Learning Generalizable and Robot-Agnostic Tool-Use Skills for a Wide Range of Tasks**". In: *Frontiers in Robotics and AI* 8, p. 380.

Jake Brawer, Aaron Hill, Ken Livingston, Eric Aaron, Joshua Bongard, and John H Long Jr (2017). "**Epigenetic Operators and the Evolution of Physically Embodied Robots**". In: *Frontiers in Robotics and AI* 4, p. 1.

Conference proceedings

Nikhil Hulle, Stéphane Aroca-Ouellette, Anthony J Ries, **Jake Brawer**, Katharina von der Wense, and Alessandro Roncone (2024). "**Eyes on the Game: Deciphering Implicit Human Signals to Infer Human Proficiency, Trust, and Intent**". In: *33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)*. Accepted for publication, proceedings pending.

Jake Brawer, Debasmita Ghose, Kate Candon, Meiying Qin, Alessandro Roncone, Marynel Vazquez, and Brian Scassellati (2023). "**Interactive Policy Shaping for Human-Robot Collaboration with Transparent Matrix Overlays**". In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE.

Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022b). "**Task-Oriented Robot-to-Human Handovers in Collaborative Tool-Use Tasks**". In: *2022 31st IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*. IEEE.

Jake Brawer, Meiying Qin, and Brian Scassellati (2020). "**A causal approach to tool affordance learning**". In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 8394–8399.

Brian Scassellati, **Jake Brawer**, Katherine Tsui, Setareh Nasihati Gilani, Melissa Malzkuhn, Barbara Manini, Adam Stone, Geo Kartheiser, Arcangelo Merla, Ari Shapiro, et al. (2018). "**Teaching Language to Deaf Infants with a Robot and a Virtual Human**". In: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, p. 553.

Zong Xuan Tan, **Jake Brawer**, and Brian Scassellati (2018). "**That's Mine! Learning Ownership Relations and Norms for Robots** ". In: *Thirty-second AAAI conference on artificial intelligence*.

Workshop proceedings

Jake Brawer, Kayleigh Bishop, Bradley Hayes, and Alessandro Roncone (2023). "**Towards A Natural Language Interface for Flexible Multi-Agent Task Assignment**". In: *Proceedings of the AAAI Symposium Series*. Vol. 2. 1, pp. 167–171.

Technical and scientific skills

Programming	**Python** (and scientific tools), **C-C++**, **LaTeX**, **Git**, **Jekyll**, **Emacs**, **Continuous integration** (with Travis), Docker
ML/AI Tools	Scikit-learn, Pytorch, NLTK, ROS, Gurobi
Robots	6+ years experience with **Baxter research robot**, 3+ years experience with **UR5 robot**
System	10+ years of daily **Linux** experience

Awards and Fellowships

July 2023, 2024, 2025	I was awarded an **Army Educational Outreach Program (AEOP) fellowship** to support my postdoctoral research 2023, 2024, and 2025 at University of Colorado, Boulder.
March 2023	**Best paper (technical track):** Our paper "Interactive Policy Shaping for Human-Robot Collaboration" won the best paper award in the technical track at the 2023 International Conference on Human-Robot Interaction (HRI 2023).

Workshops

March 2023 **Human-Machine Teaming Paradigm Workshop**
Jake Brawer, Bradley Hayes, Alessandro Roncone

March 2018 **Bridging the Gap: An NSF Workshop on Conversational Agents and Human-Robot Interaction**
Justine Cassell, Brian Scassellati, Jake Brawer, Michael Madaio

NSF Cyber–Human Systems (CHS), Robust Intelligence, National Robotics Initia-

tive. Award #1829237

Teaching Experience

Spring 2024 **CSCI 4302/5302 "Advanced Robotics"**

Lecturer. Topics covered included motion planning, mapping, localization, and reinforcement learning, culminating in a final project.

Spring 2018/2019 **Intelligent Robotics**

Teaching Assistant

Fall 2017 **Object Oriented Programming**

Teaching Assistant

Fall 2014/2015 **Perception and Action**

Teaching Assistant

Service

Leadership: Area Chair, International Conference on Human-Agent Interaction (**HAI**; 2024)

Conference reviews:
Robotics: Science and Systems (**RSS**; 2025)
International Conference on Humanoid Robots (**Humanoids**; 2018)
International Conference on Intelligent Robots and Systems (**IROS**; 2020)
International Conference on Human–Robot Interaction (**HRI**; 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2025 - **special mention for Outstanding Review (2023)**)
International Conference on Robotics and Automation (**ICRA**; 2019)

Journal reviews:
International Journal of Robotics Research (**IJRR**; 2024)
IEEE Transaction on Robotics (**T-RO**; 2024)
ACM Transactions on Human–Robot Interaction (**THRI**; 2019, 2020)

Students supervised:
Kevin Choi (2018)
Acshi Haggenmiller (2016-2017)
Sarah Widder (2017-2019)
Tan Zong Xuan (2017-2018)
Kaleb Bishop (2017-2019, 2023)
John Dallard (2021-2022)